

4.4 Option 2: Cold Environments – Landscapes and Change

Cold environments Cold environments include glacial uplands, high latitude ice-bound regions and periglacial areas. The distribution of these regions has changed significantly during the Quaternary geological period, and continues to change today. Climate determines the location of cold environments, and climate cycles have influenced the location of cold environments in the past. Much of the landscape richness in the British Isles and elsewhere is a result of past and present geomorphological processes operating in cold environments; present day study of the landscape can reveal evidence of past processes, and the same processes can be found forming landscapes today.

Management and protection Cold environments present humans with both challenges and opportunities, in the form of hazards and resources. It is increasingly recognised that cold environments are under threat from human actions and require management and protection.

Global synoptic context

Key linkages

Places

Where are cold environments today?

People

How populated are cold environments, and by whom?

Power

Who is involved in managing the threats facing cold environments today?

Suggested teaching and learning

Mapping the distribution of cold environments towards the poles, and in mountainous areas.

Using the internet to search for images of indigenous people and economic activities in cold environments.

Briefly examining an area, such as Antarctica, to investigate who manages it.

1 Defining and locating cold environments

Enquiry question: *What are cold environments and where are they found?*

What students need to learn	Suggested teaching and learning
■ Cold environments, glacial environments and periglacial environments.	■ Researching cold environments and becoming aware that not all cold environments are the same.
■ The concepts of landscape systems, glacial systems, mass balance, frequency/magnitude, equifinality and dynamic equilibrium.	■ Investigating the differences between glacial and periglacial environments and between high latitude and high altitude glacial environments.
■ The varying nature of different cold environments.	■ Investigating how the distribution of cold environments has changed over time and is still changing today.
■ The past and present day distribution of cold environments, with particular reference to the British Isles.	■ Developing awareness of the broad pattern of Britain's glacial and interglacial history and in particular the location of the glacial and periglacial environments during the most recent glacial periods of the Quaternary.

2 Climatic processes and their causes

Enquiry question: *What are the climatic processes that cause cold environments and what environmental conditions result from these?*

What students need to learn	Suggested teaching and learning
■ The climatic causes of cold environments, eg global atmospheric circulation, Polar anticyclones, the influence of latitude and altitude upon climate, day/night, seasons etc.	■ Researching the varying climatic causes of cold environments in contrasting locations.

■ Long-term global climate change leads to changes in the distribution of cold environments.	■ Weighing up past, present and future changes in global climate and how these affect cold environments.
■ The meteorological processes associated with cold climates in different types of cold environment, including extremes of temperature, winds and the nature of precipitation.	■ Investigating the significance of precipitation and wind, as well as temperature, in contributing to the distinctive features of cold climates.
■ The spatial and temporal relationships between glacial and periglacial environments.	■ Exploring the relationships between the locations of glacial and periglacial environments, both in the present and the past, with particular reference to the British Isles.

3 Distinctive landforms and landscapes

Enquiry question: *How do geomorphological processes produce distinctive landscapes and landforms in cold environments?*

What students need to learn	Suggested teaching and learning
■ The role of geomorphological processes in glacial environments. This should include both the action of ice and sub-aerial processes.	■ Examining the range of geomorphological processes that operate in different cold environments and linking these to the distinctive landscapes and landforms produced
■ The distinctive landforms produced by these processes in different glacial (upland and lowland) environments, including relict landforms produced in the past.	■ Developing an awareness that ice sheets and valley glaciers have very different landscape impacts.
■ The role of geomorphological processes in periglacial environments. This should include both above and below ground processes.	■ Investigating the differences between upland and lowland glacial landscapes and landforms.

■ The distinctive landforms produced by these processes in periglacial environments, including relict landforms produced in the past.	■ Investigating the impact of the processes during past glacial and periglacial periods when studying present-day landscapes in Britain.
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4 Challenges and opportunities

Enquiry question: *What challenges and opportunities exist in cold environments and what management issues might result from their use?*

What students need to learn	Suggested teaching and learning
■ The terms challenges and opportunities and explore the link between them.	■ Developing an awareness of the terms challenges and opportunities and understanding the link between them.
■ The challenges and opportunities of cold environments, including those of the present and the past.	■ Researching the wide range of challenges (relief, climate, avalanches, surges, meltwater floods, ground conditions, fragile ecosystems etc) and opportunities (tourism, HEP, water supply, mineral resources, etc) in different cold environments.
■ Cold environments in which humans are attempting to take the opportunities available by overcoming the challenges. These should include both glacial and periglacial environments.	■ Investigating how changes in demand and technology could lead to challenges being overcome from examples of a range of cold environments.
■ The effectiveness of different approaches to using and managing cold environments, considering the attitudes of different groups (residents, government, pressure groups, business/industry) and the conflicts that can exist between them.	■ Researching the values and attitudes of the different interest groups involved in the use and management of cold environments and examining the different approaches that can be taken to management; such as protection/conservation and sustainable management, should it be deemed necessary.